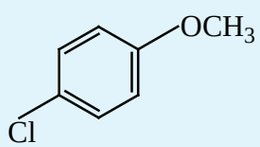
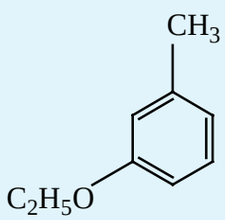
n-heptyl phenyl ether $C_6H_5-O-(CH_2)_6CH_3$ 1-phenoxyheptane	1-phenoxy	hept	ane	–
Isopentylphenylether C_6H_5 - O - CH_2 - CH_2 - CH(-CH_3) - CH_3 3-methyl-1-butoxy benzene	3-methyl-1-butoxy	benzene	–	–
Dimethylglycolate $CH_3-O-CH_2-CH_2-OCH_3$ 1,2-dimethoxyethane.	1,2-dimethoxy	eth	ane	–

Evaluate yourself

Give the IUPAC name for the following ethers and classify them as simple or mixed.

(i) $CH_3-CH_2-O-(CH_2)_3-CH_3$

(ii) 

(iii) 

(iv) $(CH_3)_3C-O-C(CH_3)_3$

(v) $CH_2=CH-CH(Cl)-O-CH_3$

(vi) dibenzyl ether

(vii) vinyl allyl ether

Preparation of ethers:

1. Inter molecular dehydration of alcohol.

We have already learnt that when ethanol is treated with con.H_2SO_4 at 443K, elimination takes place to form ethene. If the same reaction is carried out at 413K, substitution competes over elimination to form ethers.

$$2CH_3\text{-}CH_2\text{-}OH \xrightarrow[413K]{H_2SO_4} CH_3\text{-}CH_2\text{-}O\text{-}CH_2\text{-}CH_3$$

ethanol — diethylether

Mechanism:

This method is useful for the preparation of simple ethers and not suitable for preparing mixed ethers. If a mixture of two different alcohols is used, mixture of different ethers will be formed and they are difficult to separate.

XII U11-Hydroxy compounds.indd 134 11-01-2021 15:45:02